Assignment 2: Literature Search, Screening and Evidence Review (CPMG)

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1. Title, problem and review questions

Course	Methods in Business Research (MG 3010), Assignment 2
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Topic (from Assignment 1)	Contextual Provenance Defense for Persistent Memory Poisoning in LLM Agents (CPMG)
Searches run	7 November 2026

Selected problem (unchanged from Assignment 1). LLM agents store content from email, documents, websites and tool outputs in persistent memory and reuse it in later sessions. An instruction hidden in that content can be stored today and change an answer or a tool call weeks later. Assignment 1 asked whether this context-dependent memory risk can be detected at practical cost while keeping an immutable record of origin and a separate authorisation check for consequential actions.

Review questions.

RQ	Question	Key terms	Rationale	Lead
RQ1	Which attacks on the persistent memory of LLM agents have been studied, at which lifecycle stage do they act, and do they test multi-record (compositional) or delayed (dormant) activation?	*Persistent memory*: any store an agent writes and later retrieves across steps or sessions (long-term memory, experience store, RAG knowledge base). *Compositional*: harm appears only when several records combine. *Dormant*: harm is triggered by a later query.	The Assignment 1 gap rests on the claim that compositional and dormant attacks are poorly covered.	Shahmeer Ali
RQ2	Which defences protect agent memory, at which stage do they act (write, retrieval, action, repair), and what context do they use (source or provenance, the later query or action, related records)?	*Provenance*: a record of where a memory came from that the model cannot edit. *Stage*: the point in the write-store-retrieve-act lifecycle where the defence decides.	CPMG claims that no existing defence combines later context, related records and authenticated origin at low cost.	Mohammad Soban
RQ3	How are these attacks and defences evaluated (datasets, metrics, baselines, utility and cost), and which limitations do the authors report?	*Utility*: task performance on benign inputs. *Cost*: latency, tokens or extra LLM calls.	Assignment 1 proposes lifecycle metrics, detection at low false-positive rates and cost; we need to know whether this is common practice.	Both (joint)

2. Search framework and information sources

We used **PICOC**. The studies compare defences and report outcomes (the PICO part), and the setting matters: we want attacks on stores an agent writes and reuses across sessions, not one-shot prompt attacks on a chatbot. The Context element keeps the search on that setting.

Element	Our definition
Population / Problem	LLM-based agents (single or multi-agent) that write and reuse persistent memory or retrieval stores, exposed to poisoned or injected content
Intervention / Method	Defences against memory poisoning and injection: write-time filtering, retrieval-time detection, provenance and authority controls, consensus checks, audit and repair
Comparison	No defence; pointwise prompt-injection guards and keyword filters; LLM judges or consensus checks; access-control-only designs
Outcome	Attack success rate, poison storage and activation rate, detection quality (F1, AUROC, TPR at a fixed FPR), benign utility and false rejection, cost (latency, tokens, LLM calls)
Context	Agentic assistants that ingest mixed-trust content (email, web pages, documents, tool outputs, other agents) across sessions

Information sources. Five sources were searched on 7 November 2026; a sixth (Scopus) was attempted.

Source	Role	Access and export	Status
IEEE Xplore	Publisher database (journals, conferences)	Free search; RIS export with abstracts	Searched
ACM Digital Library	Publisher database	Basic Edition: export is Premium-only, so the results page was copied (titles, venues, DOIs)	Searched
ScienceDirect	Elsevier journals	RIS export with abstracts	Searched

Source	Role	Access and export	Status
SpringerLink	Springer journals, proceedings, books	CSV export (no abstracts)	Searched
Google Scholar	Discovery and citation chasing only	No export; first 50 results by relevance copied	Searched
Scopus	Indexing database	FAST-NUCES has no institutional subscription; Elsevier returned "We cannot find a way to give you access via National University of Computer and Emerging Sciences"	Not searchable
Other methods	Assignment 1 studies, backward citation searching, and checks of ACL Anthology, PMLR, NeurIPS and ICLR proceedings	Venue and peer-review status verified on the publisher or proceedings page	Used

Peer-reviewed papers were separated from preprints by checking every candidate on its publisher or proceedings page, not on a search snippet. arXiv records are labelled as preprints; a later accepted version is cited from its venue.

3. Search terms and search strings

Term lists. Concepts C1 to C3 form the database strings. C4 and C5 were left out of most strings on purpose: outcome and defence words often appear only in a paper's body, and attack-only papers are needed for RQ1. They were applied at screening and extraction instead.

Concept (PICOC)	Primary terms	Synonyms and variants	Related terms	Used for
C1 Agent (Population)	"LLM agent(s)"	"LLM-based agent", "language model agents", agentic	"AI agents", "autonomous agents", multi-agent	All RQs

Concept (PICOC)	Primary terms	Synonyms and variants	Related terms	Used for
C2 Stored content (Context)	memory	"long-term memory", "persistent memory", "retrieval-augmented" (RAG)	"knowledge base", "shared memory", "experience store"	All RQs
C3 Attack (Problem)	"memory poisoning"	poisoning, "memory injection", "prompt injection"	backdoor, "knowledge corruption", "indirect prompt injection"	RQ1
C4 Defence (Intervention)	defense	defence, detect*, guard*	provenance, "access control", repair, audit	RQ2 (Scholar ranking, screening)
C5 Evaluation (Outcome)	"attack success rate"	ASR, AUROC, "false positive rate"	utility, latency, tokens, benchmark	RQ3 (extraction)

Strings as run (7 November 2026). Concepts are joined with AND and synonyms with OR; no NOT operator was used. Each string was adapted to the database's fields and limits.

Database	Exact string	Field and filters	Records	Change and reason
IEEE Xplore	("LLM agent" OR "LLM agents" OR "language model agents" OR agentic) AND (memory OR "retrieval-augmented") AND (poisoning OR "memory injection" OR "prompt injection")	Command Search, All Metadata; all results dated 2025 to 2026, so no year filter was needed	63	None

Database	Exact string	Field and filters	Records	Change and reason
ACM Digital Library	Same as IEEE	Advanced Search, each term in Abstract; E-Publication Date 2023 to 2027	13	None
ScienceDirect	("LLM agents" OR "language model agents") AND ("memory poisoning" OR "memory injection" OR "prompt injection")	Title, abstract or author-specified keywords; Years 2023 to 2026	10	First run used the full-text box and returned 121 mostly off-topic records; moved to the title-abstract-keywords field. ScienceDirect allows no truncation and at most 8 Boolean connectors, so the agent concept was shortened
SpringerLink	"LLM agents" AND ("memory poisoning" OR "memory injection" OR "prompt injection")	Full text (no abstract field exists); all results dated 2023 to 2027 (2027 = issue year of online-first papers)	188	None
Google Scholar	"LLM agents" ("memory poisoning" OR "memory injection" OR "prompt injection") defense	Custom range 2023 to 2026, sorted by relevance; about 4,450 hits, first 50 screened	50	Capped at 50 because Scholar has no export and relevance falls off quickly

Database	Exact string	Field and filters	Records	Change and reason
Scopus (planned)	TITLE-ABS-KEY(("LLM agent*" OR "language model agent*" OR "LLM-based agent*" OR agentic) AND (memory OR "knowledge base" OR "retrieval-augmented") AND (poison* OR "memory injection" OR "prompt injection" OR backdoor)) AND PUBYEAR > 2022 AND LANGUAGE(english)	Not run	0	No institutional access at FAST-NUCES

Search validation. We checked whether the strings retrieved the 12 studies from Assignment 1. Only 2 were retrieved (MINJA and MPBench, both through Google Scholar). The misses are explained by coverage, not wording:

- AgentPoison, A-MemGuard and BlindGuard match the string's terms in their abstracts, but NeurIPS, ICML and ACL proceedings are not indexed by IEEE Xplore, ACM, ScienceDirect or SpringerLink.
- MemPoison, MemSecBench and TMA-NM are arXiv preprints, which these databases do not index.
- MemGPT, LoCoMo and LongMemEval are memory-utility studies with no security terms, so a security string should not retrieve them.
- PIGuard is a general prompt-injection guard without agent or memory terms.

Because adding terms would not fix venue coverage, we added a second identification route: the Assignment 1 studies, backward citation searching from them, and targeted checks of the ACL Anthology, PMLR, NeurIPS and ICLR proceedings. These records are counted separately in the PRISMA diagram.

4. Inclusion and exclusion criteria

The criteria were fixed before screening and applied to every record; a record had to meet all inclusion criteria and no exclusion criterion.

ID	Inclusion criterion	Reason
I1	LLM-based agents (single or multi-agent) that write content to a store and retrieve it later: long-term, experience, reasoning or shared memory, or the agent's RAG knowledge base	Matches the Population and Context of PICOC
I2	Proposes, evaluates or benchmarks an attack on or a defence of that stored content with memory-specific results, or is a review that analyses memory poisoning as its own category	Gives evidence for RQ1 to RQ3
I3	Peer-reviewed journal article or archival conference proceedings, verified on the publisher or proceedings page	The brief asks us to separate peer-reviewed work; we synthesise peer-reviewed evidence only
I4	Published (online) 2023 to 2026	Memory-enabled LLM agents appear from 2023
I5	Written in English	Both members can read and verify English only
I6	Method and results (for reviews: scope and taxonomy) reported in enough detail to extract	Needed for the extraction table

ID	Exclusion criterion	Reason
E1	Prompt injection or jailbreak within one session, with nothing stored or retrieved later	Not persistent memory
E2	Memory or RAG work on utility, efficiency or privacy leakage only, with no integrity attack or defence	Different outcome (utility or confidentiality)
E3	Extended abstracts of 2 pages or less, posters, tutorials, position papers, talks, books, chapters, reference entries	Too little method and result detail
E4	Not peer-reviewed or not archival: arXiv and preprints.org preprints, non-archival workshop papers, theses, reports, blogs	Fails I3

ID	Exclusion criterion	Reason
E5	LLMs used as security tools (intrusion detection, penetration testing, forensics) rather than as the attacked system	Wrong direction of the problem
E6	Not about LLM agents: human or social memory, classical ML poisoning, RAG question answering without an agent	Outside the Population
E7	Memory poisoning only listed among other threats, with no memory-specific analysis or results	Nothing to extract for the RQs
E8	Duplicate or earlier version of an included study	Avoids double counting

Worked examples.

Case	Paper	Decision and reason
Included	Karamchandani et al., "Your Agent's Memories Are Not Its Own: Forged Reasoning Attack on LLM Agent Memory and Its Defense", IEEE CNS 2026	Agents with persistent memory (I1); proposes the FARMA attack and the SENTINEL defence with memory-specific attack success and false-positive results (I2); IEEE conference proceedings (I3); 2026 (I4); method and results reported (I6). Include
Excluded	An et al., "IPIGuard: A Novel Tool Dependency Graph-Based Defense Against Indirect Prompt Injection in LLM Agents", EMNLP 2025	Peer-reviewed agent-security defence, but it constrains tool calls within one task; nothing is stored and reused across sessions. Exclude under E1 at title and abstract
Borderline	Choi et al., "Membrane: A Self-Evolving Contrastive Safety Memory for LLM Agent Defense", EMNLP 2026	It keeps a contrastive memory of attack and benign pairs and reports that this memory stays stable under poisoning, which is close to CPMG's design. Its subject, however, is jailbreak detection; poisoning is tested only against the guard's own memory, not the agent's. Excluded under E7 and kept as a design reference, not as evidence

A second on-topic record, Yan's "Poisoning LLM Agent Memory via Web Content" (ACM-TURC 2026), was excluded under E3 because it is a two-page extended abstract.

5. Search results and duplicate removal

The database searches returned 324 records; 4 were duplicates, leaving 320 unique records for screening. A further 20 records came from other methods.

Source	Records	Notes
IEEE Xplore	63	RIS with abstracts
ACM Digital Library	13	Results page copied (titles, venues, DOIs, abstract snippets)
ScienceDirect	10	RIS with abstracts (refined run)
SpringerLink	188	CSV, titles only
Google Scholar	50	First 50 of about 4,450, copied with snippets
Scopus	0	No access
Databases total	**324**	
Other methods	20	10 Assignment 1 studies not found by the strings, 10 from citation and proceedings searches

Duplicate handling. All exports were merged into one table by a documented script: records were matched first on DOI, then on a normalised title (lower case, punctuation removed). Every match was checked by eye. Four duplicates were removed:

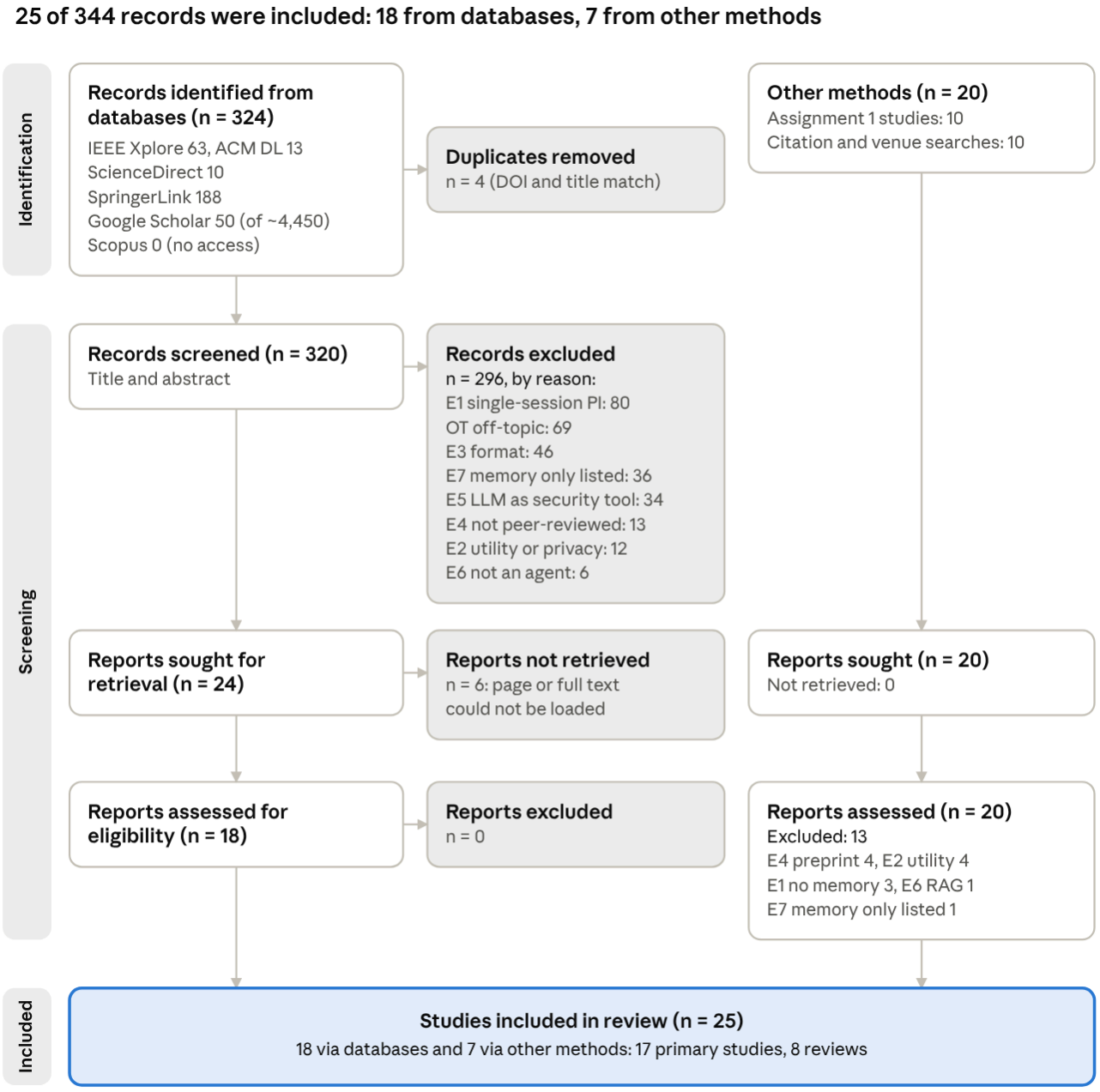
Kept record	Duplicate removed	Title
IEEE-002	GS-018	Temporal Dynamics of Memory Poisoning in Web3-Style LLM Agents
SD-010	GS-050	Memory poisoning attacks on retrieval-augmented LLM agents via deceptive semantic reasoning
SPR-011	GS-026	Prompt Infection: LLM-to-LLM Prompt Injection within Multi-agent Systems
SPR-186	SPR-187	Training LLM with Human Feedback (living version of the same entry)

Other-methods records were checked against the database records before being added; MINJA, MPBench and IPIGuard were already in the Scholar results and were not counted twice. The merged library is supplied as a BibTeX file.

6. Screening process and PRISMA flow diagram

Of 344 records, 25 studies were included: 18 from the database searches and 7 from other methods. All counts below are calculated by formula from the screening spreadsheet, so the diagram, the log and the sheet agree.

1. **Identification.** Exports from five databases were merged and de-duplicated (section 5).
2. **Title and abstract screening.** All 320 unique records were screened in a first pass (AI-assisted) against I1 to I6 and E1 to E8 and marked Include, Exclude or Uncertain with a reason code. SpringerLink exports have no abstracts, so its 188 records were screened on title; any title that could fall in scope was marked Uncertain and passed on. Clearly unrelated records were coded OT (fails I1 to I2).
3. **Independent check.** A second reviewer screened a random 15% subset (48 records, fixed seed) without seeing the first-pass decisions. Agreement was low: 24 of 48 records (Cohen's kappa 0.03; 0.08 when Include and Uncertain are merged as "pass to full text"). Of the 24 disagreements, 18 were records the second reviewer passed although they are single-session prompt-injection work (15, E1) or preprints (3, E4); two were Include versus Uncertain, which both pass. Disagreements were resolved by re-applying the written criteria, and each resolution is recorded in the Dual_Screening sheet. No first-pass decision changed, but the low agreement shows the "persistent memory" criterion needs worked examples for screeners.
4. **Full text.** 24 database reports were sought. Six could not be retrieved: FortMem (ACM, CIKM 2026) and five SpringerLink items whose pages would not load or were paywalled. The other 18 met all criteria; peer-review status was confirmed on the publisher or proceedings page. All 20 other-methods reports were assessed directly; 13 were excluded (four preprints, four memory-utility studies, three single-session prompt-injection papers, one non-agent RAG attack, one jailbreak guard).
5. **Final selection.** 25 studies: 17 primary studies and 8 reviews.



PRISMA 2020 flow diagram · databases and other methods

Records on the left came from database searches; the right column holds the Assignment 1 studies and citation or proceedings searches, which PRISMA reports without a title/abstract stage.

PRISMA 2020 checklist (items that apply at this stage).

#	Item	Status	Where
1-2	Title; abstract	Title reported; no abstract for a coursework report	Section 1
3-4	Rationale; objectives	Reported	Section 1

#	Item	Status	Where
5	Eligibility criteria	Reported	Section 4
6	Information sources and dates	Reported	Sections 2-3, Search_Log sheet
7	Full search strategies	Reported for every source	Section 3
8	Selection process	Reported	This section
9	Data collection process	Reported	Section 7
10	Data items	Reported	Section 7, Extraction sheet
11	Study risk-of-bias (quality) assessment	Reported	Section 7, Quality sheet
12, 14, 15, 21, 22	Effect measures, reporting bias, certainty	Not applicable: narrative synthesis, no meta-analysis	-
13	Synthesis methods	Thematic synthesis by RQ	Section 8
16a-b	Selection results; excluded studies with reasons	Reported	This section, Screening sheet
17-19	Study characteristics; quality; individual results	Reported	Section 7, Extraction and Quality sheets
20, 23	Synthesis results; discussion	Reported	Section 8
24	Registration and protocol	Not registered (coursework)	-
25-26	Support; competing interests	None	-
27	Availability of data and materials	Search log, screening and extraction workbook, BibTeX library	Appendices

7. Data extraction and quality assessment

All 25 studies were extracted into one structured table; 16 rated High quality, 8 Medium and 1 Low.

Extraction. Each study has every required field: study ID, full citation, publication type and peer-review status, problem, RQs informed, method, dataset, evaluation (metrics, baselines, setup), key findings, limitations (the authors' and ours), reproducibility, quality and selection decision. Facts come from the paper; anything not found is written "Not reported". Five coded fields support the synthesis: attack stage, multi-record (compositional) harm, delayed (dormant) activation, defence stage, and the context a defence uses. An **Extraction basis** column records what was read for each study. Where only the abstract and publisher page were available, the Extraction sheet's basis column says so and that study's quality score is provisional.

Quality checklist. Nine questions, scored Yes = 1, Partially = 0.5, No = 0, with a short evidence note per score: objective (Q1), method (Q2), dataset or, for reviews, search scope (Q3), baselines (Q4), metrics (Q5), evidence (Q6), limitations (Q7), code, data or protocol availability (Q8) and ethics or privacy (Q9). Q4 and Q5 are N/A for reviews. The band is the score over applicable items: **High** at 75% or more, **Medium** 50-74%, **Low** under 50%. Scores weight how much a study counts in the synthesis; they were not used to exclude studies.

Included studies at a glance. Full fields are in the Extraction and Quality sheets.

ID	Study	Venue (year)	Type	RQ	What it contributes	Quality
S01	Chen et al., AgentPoison	NeurIPS (2024)	Attack	1, 3	Trigger-based poisoning of memory or knowledge base	High
S02	Zhang et al., ASB	ICLR (2025)	Benchmark	1, 3	Memory poisoning among 27 attacks and defences, 13 LLMs	High
S03	Dong et al., MINJA	NeurIPS (2025)	Attack	1	Query-only memory injection, 98.2% injection success	High
S04	Dasgupta, MINDSTATE	IEEE URTC (2025)	Attack	1	False memories spread through agent networks	Medium
S05	Tan and Zhang, AMIA	Findings of EMNLP (2026)	Attack	1, 2	Stealthier, cheaper query-only injection on Mem0, A-MEM, MemoryOS	High
S06	Jing et al., DSRM	Eng. Appl. of AI (2026)	Attack	1	Disguised decisions injected into the knowledge base	Medium
S07	Yazdinejad and Karimipour	IEEE Access (2026)	Measurement	1, 3	2,614 trajectories; poisoning risk is temporal	High

ID	Study	Venue (year)	Type	RQ	What it contributes	Quality
S08	Mao et al., CPMP	Automated Software Eng. (2026)	Attack	1	Multi-turn poisoning of short- and long-term memory	Low
S09	Odat et al.	IEEE ICMLA (2026)	Attack and defence	1, 2	Memory amplifies corruption in drone agents; write-time monitor	High
S10	Wei et al., A-MemGuard	ICML (2026)	Defence	2, 3	Consensus over related memories plus lesson memory	High
S11	Miao et al., BlindGuard	ACL (2026)	Defence	2, 3	Unsupervised multi-agent detector incl. memory poisoning	High
S12	Karamchandani et al., FARMA/SENTINEL	IEEE CNS (2026)	Attack and defence	1-3	Forged reasoning memory; provenance-aware write-time guard	High
S13	Dhivyasree et al., CAMS	Egyptian Informatics J. (2026)	Defence	2, 3	Five-layer memory middleware, 92.3% prevention	Medium
S14	Wang et al., VAIL	IEEE BigDIA (2026)	Defence	2, 3	Retrieval authorises inspection, not use	High
S15	Gao et al., SafeFlow	IEEE BigDIA (2026)	Defence	2	Attribution-driven runtime defence	Medium
S16	Wang et al., SecureCollaRAG	ACM WWW (2026)	Attack and defence	2, 3	Source-reliability graph against knowledge corruption	High
S17	Liu et al.	ACM ICAIDS (2026)	Defence	2, 3	Propagation graph, detection and repair in shared memory	Medium
S18	Deng et al.	ACM Computing Surveys (2025)	Review	1, 2	Agent threat taxonomy with an agent-to-memory class	Medium
S19	Yu et al.	KDD (2025)	Review	1, 2	Memory as an intrinsic module with threats and countermeasures	Medium
S20	Su et al.	IEEE TPAMI (2026)	Review	1, 2	Autonomy risks; defences work in isolation	High

ID	Study	Venue (year)	Type	RQ	What it contributes	Quality
S21	Woesle and Buettner	IEEE Access (2026)	Systematic review	3	Memory boundary under-studied in 81 papers	High
S22	Jiang et al.	IEEE CAI (2026)	Review (SoK)	1, 2	Persistent memory poisoning as a runtime supply-chain attack	Medium
S23	Sammartino	Knowledge-Based Systems (2026)	Systematic review	1-3	No prompt-level defence survives adaptive attacks	High
S24	Hu et al.	Information and Software Tech. (2026)	Systematic review	3	1 of 8 code-agent benchmarks covers memory	High
S25	Lin et al.	EMNLP (2026)	Review	1-3	Memory lifecycle framework; anchor security in storage-time provenance	High

Recency: all 25 studies are from 2022 onward and 24 from 2025 onward. The set includes foundational attacks (S01, S03) and eight reviews.

8. Thematic synthesis and research gaps

The review confirms the Assignment 1 gap: attacks now exploit memory that builds up over time, while defences check one stage with one kind of context, and their evaluations are not comparable.

Attacks moved from one record to accumulated state (RQ1)

Early work showed that very little poison is needed: AgentPoison reached about 60% target-action success with under 0.1% poisoned records [S01], and MINJA did it through ordinary queries [S03]; AMIA then cut the query cost [S05]. Newer studies attack state that accumulates: forged reasoning traces that cite each other [S12], chains and slow drift across a trajectory [S07], multi-turn concealment [S08] and spread across agents that share memory [S04, S17]. Seven of the 17 primary studies involve harm that needs several records or agents, and five involve delayed activation. The two measurement studies explain why this matters. Attacks stay indistinguishable from benign behaviour until late [S07], and bounded memory raises mission failure from 31.5% to 53.3% after a single corrupted input [S09]. Both conclude that step-by-step or outcome-only evaluation undercounts harm.

Defences act at one stage with one kind of context (RQ2)

Nine primary studies propose or test defences. Three act at write time [S09, S12, S13], three at retrieval or use [S10, S14, S16], two on agent communication [S11, S15], and three add repair [S10, S13, S17]. Six use source or provenance signals and six use related records, but only two look at the moment of use [S10, S14]. None combines all three in one model. A-MemGuard uses the query and related memories but needs several LLM calls and ignores origin [S10]. SENTINEL checks provenance in under 1 ms per write, yet a paraphrasing adaptive attacker bypassed it [S12]. Graph detectors combine source reliability with neighbouring records but not the later action [S16, S17]. CAMS comes closest by stacking five separate layers, but its abstract reports only a prevention rate on two attack sets [S13].

Defence	Stage	Source or provenance	Related records	Moment of use	Cost reported
S09 Memory Contract Monitor	Write	-	-	-	No
S10 A-MemGuard	Retrieval + repair	-	Yes	Yes (query)	Appendix
S11 BlindGuard	Communication	-	Yes (neighbours)	-	No
S12 SENTINEL	Write	Yes	Yes (self-reference)	-	Yes
S13 CAMS	Write + re-scan	Yes	Yes	-	No
S14 VAIL	Use	Yes (ledger)	-	Yes (live re-check)	No
S15 SafeFlow	Runtime	Yes (attribution)	-	-	No
S16 SecureCollaRAG	Retrieval	Yes	Yes (graph)	-	No
S17 Liu et al.	Detection + repair	Yes	Yes (graph)	-	No

The reviews add a consistent message: robustness should come from the system around the model. They call for storage-time provenance [S25], cryptographic provenance rather than semantic inference [S22] and limits on what the system may do [S23], and note that existing defences work in isolation [S20]. VAIL puts this into practice by letting retrieval authorise inspection but not use [S14].

Evaluations are not yet comparable (RQ3)

Attack success or failure rate is the main outcome in most primary studies, but detection is reported as AUC or a raw false-positive rate, and none of the studies we read reports detection at a fixed low false-positive rate. Seven of 17 report benign utility; SENTINEL is the only defence we found reporting latency, and A-MemGuard leaves token cost to an appendix. There is no shared memory-poisoning benchmark: three studies reuse EHRAgent and StrategyQA [S01, S10, S12], while six build their own testbeds. Adaptive attackers appear in only two studies [S12, S16], which matches the finding that no prompt-level defence survives adaptive attack [S23]. The systematic reviews see the same imbalance field-wide: research concentrates on the prompt interface while memory boundaries remain under-characterised [S21], and one of eight code-agent benchmarks covers memory [S24].

Strengths and weaknesses follow venue type. Main-track papers usually have baselines, repeated trials and public code (16 rated High). Short conference papers and single in-house testbeds account for most Medium and Low ratings.

Research gaps supported by several studies

1. **Multi-record and delayed attacks are measured, but rarely defended.** Defences are tested mostly on single-record attacks such as AgentPoison and MINJA [S10, S13]; delayed activation is characterised [S07] but not used as a defence test.
2. **No low-cost defence combines the later query or action, related records and authenticated origin** [S10, S12, S13, S16, S17].
3. **No common evaluation:** no fixed low-FPR operating points, little cost reporting, no shared benchmark and few adaptive attacks [S12, S21, S23, S24].
4. **Origin-bound authority is recommended but not tested with a content detector** in peer-reviewed work [S14, S22, S25].

How this changes the Assignment 1 problem

The evidence **confirms** the gap on compositional and dormant attacks and on combining context with provenance. It **sharpens** the evaluation plan: report TPR at 1% and 5% FPR and cost, add a paraphrasing adaptive attacker and forged reasoning traces [S12], and treat shared multi-agent memory [S04, S17] as a held-out family. It **changes** the evidence base: Assignment 1 leaned on preprints (MPBench, MemPoison, MemSecBench, TMA-NM). We now ground the gap in peer-reviewed studies, label MPBench as a non-archival workshop benchmark, and add peer-reviewed testbeds (EHRAgent and StrategyQA settings from S01, S10, S12; ASB [S02]) for comparability.

9. Limitations of the search and selection process

The main limits are database access and our peer-reviewed-only rule, both of which may under-represent the newest memory-poisoning work.

- **Coverage.** Scopus was not accessible and Web of Science was not available to us. The searched databases do not index NeurIPS, ICML, ICLR or ACL proceedings, so key studies came through other methods, which leans toward work we already knew.
- **Google Scholar cap.** Only the first 50 of about 4,450 results were screened; relevant papers further down may be missed.
- **Incomplete records.** SpringerLink exports have no abstracts, so 188 records were screened on titles; ACM export needed a paid edition; six reports could not be retrieved.
- **Full-text depth.** Several extractions rely on the abstract and publisher page; these rows are flagged and quality scores for them are provisional.
- **Peer-review rule.** Excluding preprints and non-archival workshop papers removed recent studies on sleeper, compositional and lifecycle attacks (for example MemPoison, MemSecBench and the ICML 2026 workshop papers). The gap on these attacks may therefore be smaller than our evidence suggests.
- **Screening reliability.** All records had one AI-assisted first pass; a second reviewer checked a 15% subset and agreement was low (kappa 0.03), so borderline decisions depend on how "persistent memory" is read. Full-text decisions and quality scores rest on group judgement.
- **Scope.** English only, 2023 to 2026, searches run on a single day (7 November 2026); some 2026 proceedings metadata may still change.

10. Contribution statement, references and appendices

This is a two-member group submission. The table records who did what; both members agreed the final versions of all deliverables.

Member	Research questions	Tasks
Shahmeer Ali (23I-0119)	Lead on RQ1 (attacks); joint on RQ3 (evaluation)	Search strategy and strings; ran and exported all five database searches on 7 November 2026; deduplication; first-pass title/abstract and full-text screening; data extraction and quality appraisal; PRISMA diagram and checklist; reference library; drafted the synthesis.
Mohammad Soban (23I-0056)	Lead on RQ2 (defences); joint on RQ3 (evaluation)	Independent second screening of the 48-record subset used for the agreement check (section 6); Dual_Screening sheet in the workbook.

References (IEEE style).

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Appendices (submitted files).

- **A. Search log**: Search_Log sheet of the workbook *A2_Screening_and_Extraction.xlsx*, with the exact strings, fields, filters and counts.
- **B. Screening spreadsheet**: Screening sheet (all 344 records, title/abstract and full-text decisions with reasons) and Dual_Screening sheet (48-record independent check with agreement and kappa formulas).
- **C. PRISMA counts**: PRISMA_Counts sheet; every number in the diagram is a formula over the Screening sheet, with built-in checks.
- **D. Extraction and quality tables**: Extraction and Quality sheets for S01-S25.
- **E. Reference library**: *A2_reference_library.bib* (all 344 records, duplicates flagged) and *A2_included_studies.bib* (25 included studies).
- **F. Full texts**: DOI or publisher link for every included study in the Extraction sheet.